20261001 Quiz

for (MATH102) Calculus II

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Problem 1.5. Find a linear function $\ell\colon \mathbb{R}^2 \to \mathbb{R}$ satisfying

$$\ell(1,2) = 3, \qquad \ell(2,3) = 5.$$

Solution. By Theorem 1.2, every linear function $\ell\colon \mathbb{R}^2 \to \mathbb{R}$ has the form

$$\ell(x,y) = px + qy$$

for some $p, q \in \mathbb{R}$. The given values yield

$$p + 2q = 3, \qquad 2p + 3q = 5.$$

Thus $p = q = 1$, and hence

$$\boxed{\ell(x,y) = x + y}.$$

□

Problem 1.6. Let $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$, and $\mathbf{w} = (w_1, w_2)$ be vectors in $\mathbb{R}^2$, and let $a, b, c \in \mathbb{R}$. Using the definitions

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2), \qquad c\mathbf{u} = (cu_1, cu_2), \qquad -\mathbf{u} = (-u_1, -u_2),$$

show the following properties.

(a) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
(b) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$.
(c) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
(d) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$.
(e) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

Solution. Each identity follows componentwise from the definitions.

(a)
$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2) = (v_1 + u_1, v_2 + u_2) = \mathbf{v} + \mathbf{u}.$$

(b)
$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (u_1 + v_1 + w_1, u_2 + v_2 + w_2) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}.$$

(c)
$$c(\mathbf{u} + \mathbf{v}) = (cu_1 + cv_1, cu_2 + cv_2) = c\mathbf{u} + c\mathbf{v}.$$

(d)
$$(a + b)\mathbf{u} = (au_1 + bu_1, au_2 + bu_2) = a\mathbf{u} + b\mathbf{u}.$$

(e)

$$\mathbf{u} + (-\mathbf{u}) = (0, 0) = \mathbf{0}.$$

□

Problem 1.16. Let

$$\mathbf{u} = (1, 3), \qquad \mathbf{v} = (3, 1).$$

(a) Determine whether $\mathbf{u}$ and $\mathbf{v}$ are linearly independent.
(b) Express $(4, 4)$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$.
(c) Express $(4, 5)$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

Solution.

(a) If $a\mathbf{u} + b\mathbf{v} = \mathbf{0}$, then

$$a + 3b = 0, \qquad 3a + b = 0.$$

These equations give $a = b = 0$. Therefore $\mathbf{u}$ and $\mathbf{v}$ are

$$\boxed{\text{linearly independent}}.$$

(b) Solving

$$a + 3b = 4, \qquad 3a + b = 4$$

gives $a = b = 1$. Hence

$$\boxed{(4, 4) = \mathbf{u} + \mathbf{v}}.$$

(c) Solving

$$a + 3b = 4, \qquad 3a + b = 5$$

gives

$$a = \frac{11}{8}, \qquad b = \frac{7}{8}.$$

Therefore

$$\boxed{(4, 5) = \frac{11}{8}\mathbf{u} + \frac{7}{8}\mathbf{v}}.$$

□

Problem 1.21. Let $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$, and $\mathbf{w} = (w_1, w_2)$. Prove

(a) the distributive property $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
(b) the commutative property $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.

Solution. Using the definition of the dot product,

(a)

$$\begin{aligned} \mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) &= u_1(v_1 + w_1) + u_2(v_2 + w_2) \\ &= (u_1 v_1 + u_2 v_2) + (u_1 w_1 + u_2 w_2) \\ &= \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}. \end{aligned}$$

(b)

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = v_1 u_1 + v_2 u_2 = \mathbf{v} \cdot \mathbf{u}.$$

□

Problem 1.25. Let $\ell\colon \mathbb{R}^2 \to \mathbb{R}$ be linear and satisfy

$$\ell(2,1) = 3, \qquad \ell(1,1) = 2.$$

Find a vector $\mathbf{c} \in \mathbb{R}^2$ such that

$$\ell(\mathbf{u}) = \mathbf{c} \cdot \mathbf{u}$$

for every $\mathbf{u} \in \mathbb{R}^2$.

Solution. By Theorem 1.3, there is a vector $\mathbf{c} = (p, q)$ such that $\ell(\mathbf{u}) = \mathbf{c} \cdot \mathbf{u}$ for every $\mathbf{u} \in \mathbb{R}^2$. The given values yield

$$2p + q = 3, \qquad p + q = 2.$$

Thus $p = q = 1$, so

$$\boxed{\mathbf{c} = (1,1)}.$$

□

Problem 1.26. Find the cosine of the angle between the vectors

$$\mathbf{u} = (1,2), \qquad \mathbf{v} = (3,1).$$

Solution. By Theorem 1.5,

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

Here

$$\mathbf{u} \cdot \mathbf{v} = 5, \qquad \|\mathbf{u}\| = \sqrt{5}, \qquad \|\mathbf{v}\| = \sqrt{10}.$$

Therefore

$$\boxed{\cos\theta = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}}.$$

□

Problem 1.41. Show that the vectors

$$(1,1,1,1), \quad (0,1,1,1), \quad (0,0,1,1), \quad (0,0,0,1)$$

are linearly independent in $\mathbb{R}^4$.

Solution. Let these vectors be $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$, respectively, and suppose

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + c_4\mathbf{v}_4 = \mathbf{0}.$$

The first coordinate gives $c_1 = 0$. The second then gives $c_2 = 0$, the third gives $c_3 = 0$, and the fourth gives $c_4 = 0$. Hence the only linear combination equal to $\mathbf{0}$ is the trivial one, so the four vectors are linearly independent.

□

Problem 1.42. Determine whether the vectors

$$(1,2,1), \quad (1,2,2), \quad (1,2,3), \quad (1,2,4)$$

are linearly dependent or linearly independent. Identify the theorem of this section that is particularly applicable.

Solution. There are four vectors in $\mathbb{R}^3$. By Theorem 1.8, any $n+1$ vectors in $\mathbb{R}^n$ are linearly dependent. Taking $n = 3$, these four vectors are therefore

$$\boxed{\text{linearly dependent}}.$$

□

Problem 1.47. Show that the vectors

$$(1,1,1), \qquad (1,2,3), \qquad (3,2,1)$$

are linearly dependent.

Solution. Let

$$\mathbf{v}_1 = (1,1,1), \qquad \mathbf{v}_2 = (1,2,3), \qquad \mathbf{v}_3 = (3,2,1).$$

We find the nontrivial relation

$$-4\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}.$$

Hence the vectors are linearly dependent.

□

Problem 1.62. Let

$$\mathbf{c} = (c_1, \ldots, c_n) \in \mathbb{R}^n, \qquad \mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n.$$

Show that the function

$$f(\mathbf{x}) = x_1 + \mathbf{c} \cdot \mathbf{x}$$

is linear by finding $\mathbf{d} = (d_1, \ldots, d_n)$ such that

$$f(\mathbf{x}) = \mathbf{d} \cdot \mathbf{x}.$$

Solution. Since

$$f(\mathbf{x}) = (1 + c_1)x_1 + c_2x_2 + \cdots + c_nx_n,$$

take

$$\boxed{\mathbf{d} = (1 + c_1, c_2, \ldots, c_n) = \mathbf{e}_1 + \mathbf{c}}.$$

Then $f(\mathbf{x}) = \mathbf{d} \cdot \mathbf{x}$. The dot product with a fixed vector is linear in $\mathbf{x}$, so f is linear.

□

Problem 1.84. Evaluate each of the following products.

(a)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

(b)

$$\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 6 \\ 6 \end{pmatrix}.$$

(c)

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

(d)

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^2.$$

Solution. Using the definition of matrix multiplication,

(a)
$$\boxed{\begin{pmatrix} x_1 \\ 2x_2 \\ 6x_3 \end{pmatrix}}.$$

(b)
$$\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} 6 \\ 6 \end{pmatrix} = \boxed{\begin{pmatrix} -6 \\ 42 \end{pmatrix}}.$$

(c)
$$\boxed{\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}}.$$

(d)
$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^2 = \begin{pmatrix} \cos^2\theta - \sin^2\theta & -2\sin\theta\cos\theta \\ 2\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{pmatrix}$$
$$= \boxed{\begin{pmatrix} \cos(2\theta) & -\sin(2\theta) \\ \sin(2\theta) & \cos(2\theta) \end{pmatrix}}.$$

□

Problem 1.95. Find equations in the form
$$ax + by + cz = d$$
for the following planes in $\mathbb{R}^3$.

(a) The plane through the origin with normal vector $(1, 0, 0)$.

(b) The plane through $(0, 0, 0)$, $(0, 1, 1)$, and $(-3, 0, 0)$.

(c) The plane containing $(1, 1, 1)$ and parallel to the plane
$$x - 3y + 5z = 60.$$

Solution.

(a) A normal vector is $(1, 0, 0)$, and the plane passes through the origin. Hence
$$\boxed{x = 0}.$$

(b) Two vectors in the plane are
$$\mathbf{v} = (0, 1, 1), \qquad \mathbf{w} = (-3, 0, 0).$$
Their cross product is
$$\mathbf{v} \times \mathbf{w} = (0, -3, 3),$$
so a normal vector is $(0, -1, 1)$. Since the plane passes through the origin,
$$\boxed{-y + z = 0}.$$

(c) A parallel plane has the same normal vector $(1, -3, 5)$, so it has the form
$$x - 3y + 5z = d.$$
Substituting $(1, 1, 1)$ gives $d = 3$. Therefore
$$\boxed{x - 3y + 5z = 3}.$$

□

Problem 1.104. Using the formula

$$\mathbf{v} \times \mathbf{w} = (v_2w_3 - v_3w_2,\, v_3w_1 - v_1w_3,\, v_1w_2 - v_2w_1),$$

evaluate the following cross products.

(a) $(1,0,0) \times (0,1,0)$.
(b) $(0,1,0) \times (1,0,0)$.
(c) $(0,0,1) \times (a,b,c)$.

Solution. Substituting directly into the given formula,

(a)

$$\boxed{(1,0,0) \times (0,1,0) = (0,0,1)}.$$

(b)

$$\boxed{(0,1,0) \times (1,0,0) = (0,0,-1)}.$$

(c)

$$\boxed{(0,0,1) \times (a,b,c) = (-b,a,0)}.$$

□

Problem 1.105. Using

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \qquad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \qquad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2,$$

and the distributive and anticommutative laws

$$\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}, \qquad \mathbf{v} \times \mathbf{w} = -\mathbf{w} \times \mathbf{v},$$

evaluate the following cross products.

(a) $(1,0,0) \times (0,1,0)$.
(b) $\mathbf{e}_2 \times (\mathbf{e}_1 + \mathbf{e}_3)$.
(c) $(2\mathbf{e}_1 + 3\mathbf{e}_3) \times (a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3)$.

Solution.

(a) Since $(1,0,0) = \mathbf{e}_1$ and $(0,1,0) = \mathbf{e}_2$,

$$\boxed{(1,0,0) \times (0,1,0) = \mathbf{e}_3 = (0,0,1)}.$$

(b)

$$\begin{aligned}\mathbf{e}_2 \times (\mathbf{e}_1 + \mathbf{e}_3) &= \mathbf{e}_2 \times \mathbf{e}_1 + \mathbf{e}_2 \times \mathbf{e}_3 \\ &= -\mathbf{e}_3 + \mathbf{e}_1.\end{aligned}$$

Thus

$$\boxed{\mathbf{e}_2 \times (\mathbf{e}_1 + \mathbf{e}_3) = \mathbf{e}_1 - \mathbf{e}_3}.$$

(c) Using distributivity and anticommutativity,

$$\begin{aligned}&(2\mathbf{e}_1 + 3\mathbf{e}_3) \times (a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3) \\ &\quad = 2b\mathbf{e}_3 - 2c\mathbf{e}_2 + 3a\mathbf{e}_2 - 3b\mathbf{e}_1 \\ &\quad = -3b\mathbf{e}_1 + (3a - 2c)\mathbf{e}_2 + 2b\mathbf{e}_3.\end{aligned}$$

Hence

$$\boxed{-3b\mathbf{e}_1 + (3a - 2c)\mathbf{e}_2 + 2b\mathbf{e}_3}.$$

□

Problem 2.4. Show that a constant function $\mathbf{F}\colon \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if

$$\mathbf{F}(\mathbf{x}) = \mathbf{0}$$

for every $\mathbf{x} \in \mathbb{R}^n$.

Solution. Suppose first that $\mathbf{F}$ is linear. Since $\mathbf{F}$ is constant,

$$\mathbf{F}(\mathbf{x}) = \mathbf{F}(\mathbf{0})$$

for every $\mathbf{x}$. By linearity,

$$\mathbf{F}(\mathbf{0}) = \mathbf{F}(0\mathbf{x}) = 0\mathbf{F}(\mathbf{x}) = \mathbf{0}.$$

Thus $\mathbf{F}$ is the zero function.

Conversely, if $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ for every $\mathbf{x}$, then

$$\mathbf{F}(c\mathbf{x}) = \mathbf{0} = c\mathbf{F}(\mathbf{x})$$

and

$$\mathbf{F}(\mathbf{x}+\mathbf{y}) = \mathbf{0} = \mathbf{F}(\mathbf{x}) + \mathbf{F}(\mathbf{y}).$$

Hence $\mathbf{F}$ is linear.

□

Problem 2.5. The plane in $\mathbb{R}^3$ given by

$$z = 2x + 3y$$

is the 0 level set of a linear function $\ell\colon \mathbb{R}^3 \to \mathbb{R}$. Find such a function ℓ.

Solution. Rewrite the plane equation as

$$2x + 3y - z = 0.$$

Thus the linear function

$$\boxed{\ell(x, y, z) = 2x + 3y - z}$$

has exactly this plane as its 0 level set.

□

Problem 2.13. Sketch or describe the c level sets $f(\mathbf{x}) = c$ in $\mathbb{R}^3$ for the following functions and values of c. Each function is considered on its natural domain.

(a) $f(\mathbf{x}) = \|\mathbf{x}\|^2$, for $c = \frac{1}{2}, 1, 2$.

(b) $f(\mathbf{x}) = \|\mathbf{x}\|$, for $c = \frac{1}{2}, 1, 2$.

(c) $f(\mathbf{x}) = \dfrac{1}{\|\mathbf{x}\|^2}$, for $c = \frac{1}{2}, 1, 2$.

(d) $f(\mathbf{x}) = \mathbf{u} \cdot \mathbf{x}$, for $c = \frac{1}{2}, 1, 2$, where $\mathbf{u}$ is a unit vector. Hint: write $\mathbf{x}$ as the sum of a vector parallel to $\mathbf{u}$ and a vector orthogonal to $\mathbf{u}$.

Solution.

(a) The c level set satisfies $\|\mathbf{x}\|^2 = c$, so it is

$$\boxed{\{\mathbf{x} \in \mathbb{R}^3 \mid \|\mathbf{x}\| = \sqrt{c}\}}.$$

For $c = \frac{1}{2}, 1, 2$, these are spheres centered at the origin with radii $\frac{1}{\sqrt{2}}$, 1, and $\sqrt{2}$, respectively.

(b) The c level set is

$$\boxed{\{\mathbf{x} \in \mathbb{R}^3 \mid \|\mathbf{x}\| = c\}}.$$

Thus for $c = \frac{1}{2}, 1, 2$, the level sets are spheres centered at the origin with radii $\frac{1}{2}$, 1, and 2, respectively.

(c) Since

$$\frac{1}{\|\mathbf{x}\|^2} = c \iff \|\mathbf{x}\| = \frac{1}{\sqrt{c}},$$

the c level set is

$$\boxed{\left\{\mathbf{x} \in \mathbb{R}^3 \mid \|\mathbf{x}\| = \frac{1}{\sqrt{c}}\right\}}.$$

For $c = \frac{1}{2}, 1, 2$, these are spheres centered at the origin with radii $\sqrt{2}$, 1, and $\frac{1}{\sqrt{2}}$, respectively.

(d) Write

$$\mathbf{x} = t\mathbf{u} + \mathbf{y}, \qquad \mathbf{y} \perp \mathbf{u}.$$

Since $\|\mathbf{u}\| = 1$,

$$\mathbf{u} \cdot \mathbf{x} = t\|\mathbf{u}\|^2 + \mathbf{u} \cdot \mathbf{y} = t.$$

Hence the c level set is

$$\boxed{\{\mathbf{x} \in \mathbb{R}^3 \mid \mathbf{u} \cdot \mathbf{x} = c\} = \{c\mathbf{u} + \mathbf{y} \mid \mathbf{y} \in \mathbb{R}^3,\ \mathbf{y} \perp \mathbf{u}\}}.$$

For $c = \frac{1}{2}, 1, 2$, these are planes perpendicular to $\mathbf{u}$ through $\frac{1}{2}\mathbf{u}$, $\mathbf{u}$, and $2\mathbf{u}$, respectively.

□

Problem 2.35. Show that the set

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 < 1\}$$

is open.

Solution. Let $\mathbf{p} = (a, b, c) \in S$ and set

$$\rho = \sqrt{a^2 + b^2} < 1, \qquad r = 1 - \rho > 0.$$

If $\mathbf{q} = (x, y, z)$ satisfies $\|\mathbf{q} - \mathbf{p}\| < r$, then by the triangle inequality in the xy-plane,

$$\begin{aligned}\sqrt{x^2 + y^2} &\le \sqrt{(x-a)^2 + (y-b)^2} + \sqrt{a^2 + b^2} \\ &\le \|\mathbf{q} - \mathbf{p}\| + \rho \\ &< r + \rho = 1.\end{aligned}$$

Thus $\mathbf{q} \in S$, so $B_r(\mathbf{p}) \subseteq S$. Every point of S is therefore an interior point, and S is open by Definition 2.14.

□

Problem 2.36. Let

$$S = \mathbb{R}^2 \setminus \{\mathbf{0}\}.$$

Show that $\mathbf{0}$ is a boundary point of S.

Solution. Let $r > 0$. The open ball $B_r(\mathbf{0})$ contains $\mathbf{0}$, which is not in S, and it also contains, for example,

$$\left(\frac{r}{2}, 0\right) \in S.$$

Thus every ball centered at $\mathbf{0}$ contains a point of S and a point not in S. By Definition 2.15, $\mathbf{0}$ is a boundary point of S.

□

Problem 2.37. Let

$$T = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0,\ x + y \leq 1\}.$$

(a) Describe the boundary of T.

(b) Show that the point $(0.0001, 0.9998)$ is an interior point of T.

Solution.

(a) The boundary consists of the three line segments

$$\boxed{\begin{aligned}\partial T = &\{(x, y) \in \mathbb{R}^2 \mid x = 0,\ 0 \leq y \leq 1\} \\ &\cup \{(x, y) \in \mathbb{R}^2 \mid y = 0,\ 0 \leq x \leq 1\} \\ &\cup \{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0,\ x + y = 1\}.\end{aligned}}$$

(b) Let

$$\mathbf{p} = (0.0001, 0.9998), \qquad r = 0.00005.$$

If $\mathbf{q} = (x, y)$ satisfies $\|\mathbf{q} - \mathbf{p}\| < r$, then

$$|x - 0.0001| < r, \qquad |y - 0.9998| < r.$$

Therefore

$$x > 0.00005 > 0, \qquad y > 0.99975 > 0,$$

and

$$x + y < 0.0001 + 0.9998 + 2r = 1.$$

Hence $B_r(\mathbf{p}) \subseteq T$. By Definition 2.13, $\mathbf{p}$ is an interior point of T.

□